

packages/graph-core/src/analyzer/llm-analyzer.ts

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Overview

A new module `packages/graph-core/src/analyzer/llm-analyzer.ts` is added. It defines typed interfaces and helper functions for generating LLM prompts and parsing their JSON responses for both file-level and project-level analysis.

Key changes

- **Interfaces**
 - `LLMFileAnalysis` (lines 1-9) – shape for file analysis.
 - `LLMProjectSummary` (lines 10-18) – shape for project summaries.
- **Prompt builders**
 - `buildFileAnalysisPrompt` (lines 19-42) – includes file content, project context, and requires a JSON response.
 - `buildProjectSummaryPrompt` (lines 48-75) – lists all files, optional sample snippets, and demands a JSON object.
- **JSON extraction**
 - `extractJson` (lines 81-95) – removes markdown fences or raw JSON blocks.
- **Response parsers**
 - `parseFileAnalysisResponse` (lines 102-143) – validates fields, normalizes `complexity` using `VALID_COMPLEXITIES` (line 97), returns `null` on failure.
 - `parseProjectSummaryResponse` (lines 148-186) – validates description, frameworks, and layers with a type guard; returns `null` on failure.

Impact

- Typed contracts and defensive parsing reduce runtime errors from malformed LLM output.
- Centralized prompt logic simplifies future adjustments; interfaces keep the shape explicit.
- No existing exports are modified; current consumers remain unaffected.
- No new dependencies; operations are lightweight string manipulations.

Risks & follow-ups

- `extractJson` may fail if the response lacks fences or contains extraneous text – test with varied LLM outputs.
- `VALID_COMPLEXITIES` covers only "simple", "moderate", "complex"; consider a fallback for unexpected values.
- Layer filtering in `parseProjectSummaryResponse` relies on a type guard – test with malformed layer objects to confirm graceful degradation.

- Run lint, unit tests, and the production build to catch any type mismatches introduced by the new interfaces.